JobPilot

Smart Job Matcher -- Technical Brief

BAX-423 Big Data | Spring 2026 | Dr. Rahul Makhijani | UC Davis MSBA

1. System Architecture

JobPilot is a single-process Flask application that ingests real job postings, matches them to an uploaded resume using hybrid sparse + dense retrieval, re-ranks candidates through a multi-stage pipeline, adapts to user feedback via the Rocchio algorithm, and lets users download results or generate a tailored resume.

Layer	Technology	Purpose
Web Framework	Flask 3.1 + Gunicorn	HTTP routing, sessions, template rendering
Database	SQLite + SQLAlchemy	Jobs, sessions, feedback -- zero-setup file DB
Dense Retrieval	FAISS IndexFlatIP	ANN search over 8,280 job vectors (384-dim)
Sparse Retrieval	BM25Okapi (rank-bm25)	Keyword candidate generation, top-500/query
Embedding Model	BAAI/bge-small-en (ONNX)	384-dim vectors via fastembed; no PyTorch
Adaptive Learn.	Rocchio Algorithm	Query vector updated from accept/reject signals
Streaming	APScheduler + Adzuna API	Background polling; SHA-256 deduplication
Resume Gen.	Claude API / Template fallback	Tailors resume text to selected job description
Deployment	Render (Docker)	https://jobpilot-fhpv.onrender.com

Pipeline Flow

PDF upload -> pdfplumber extraction -> 384-dim query vector -> Stage 1: hard filters (dealbreakers, salary, seniority, experience cap, US-only, no-contract) -> Stage 2: BM25 top-500 + FAISS ANN top-500 merged -> Stage 3: hybrid re-rank (0.7 x dense + 0.3 x BM25 + 0.1 x skill overlap) -> Rocchio update after each feedback round

2. BAX-423 Technique Choices & Benchmark Results

Technique 1 -- BM25 (Sparse Retrieval) [Text Mining / Information Retrieval lectures]

* TF-IDF probabilistic ranking over job title + company + skills + description[:300].

* Why cho

Technique 2 -- Dense Embeddings + FAISS ANN [NLP / Embeddings lectures]

* BAAI/bge-small-en-v1.5 (ONNX via fastembed) encodes jobs at index time to 384-dim unit vectors.

* FAISS In

Benchmark (Recall@10 and NDCG@10 -- 3 test profiles on 8,280 real Kaggle jobs)

Ground truth: jobs matching user skills OR role keywords in title/description.

Method	ML Eng	ML Eng	Data Anlst	Data Anlst	MLOps	MLOps
	R@10	N@10	R@10	N@10	R@10	N@10
BM25 (sparse)	0.013	0.854	0.001	1.000	0.008	1.000
SBERT+FAISS (dense)	0.006	1.000	0.001	1.000	0.003	1.000
Hybrid (production)	0.006	0.605	0.001	0.943	0.004	0.784

Dense retrieval achieves NDCG@10 = 1.000 -- relevant jobs ranked at position 1. BM25 provides broader recall. Hybrid pipeline used in production (0.7 x dense + 0.3 x BM25).

3. Adaptive Learning -- Rocchio Algorithm

After each accept/reject/skip round, the Rocchio algorithm updates the query vector in 384-dimensional embedding space:

```
q_new = a*q_orig + b*mean(accepted) - g*mean(rejected)   [a=1.0, b=0.8, g=0.1]
```

* Each accepted/rejected job is encoded to 384-dim and shifts the query vector.

* Updated

4. Data Pipeline

* Source: Techmap 'international-job-postings-september-2021' (Kaggle). 10,000 records sampled; 8,280 unique after SHA-256 deduplication (17.1% dedup rate).

* Techr
orgAddress.

5. Test Persona Results

Persona	Key Constraints	Strong	Partial	Irrel.	Top-10 Relevant	Explain (3 jobs)	Resume (2 jobs)	PASS?
Aisha (ML Pivoter)	No senior/staff titles No defense/military Remote/Bay Area, $140k+	2	8	0	10/10 PASS	3/3 PASS	2/2 PASS	PASS
Marcus (New Grad)	No 3+ yr exp required No contract/temp roles Any US, $80k+	3	7	0	10/10 PASS	3/3 PASS	2/2 PASS	PASS
Priya (Sr. MLOps)	No junior titles NYC/Remote, $200k+ No small companies	1	8	1	9/10 PASS	3/3 PASS	2/2 PASS	PASS
Kenji (Research/Visa)	No contract/temp (visa) US only, $120k+ Research focus	3	2	5	5/10 NOTE*	3/3 PASS	2/2 PASS	PASS

** Kenji ranks 5-10 show noise due to 2021 Kaggle dataset's limited research/AI postings. Top-3 include Sr. ML Engineer (100% semantic match) and AI Engineer -- both direct strong matches. No contract roles confirmed. US-only filter active.*

Capability Coverage -- All 6 Required Capabilities

#	Capability	Verification
1	Job Data Ingestion & Streaming	8,280 real Kaggle jobs; 1,711 dupes removed; Adzuna streaming
2	Profile Intake & Skill Matching	PDF + form upload; skills extracted from description text; filters active
3	Embedding-Based Retrieval	BAAI/bge-small-en -> FAISS IndexFlatIP; 93-100% semantic scores in QA
4	Multi-Stage Ranking + Metric	Hard filters -> BM25+FAISS -> hybrid; NDCG@10 benchmarked, shown in
5	Adaptive Learning	Rocchio; NDCG 0.500->1.000 after 1 round; round counter + metrics in
6	Download Top Jobs	CSV: 51 rows, title/company/location/salary/URL/description -- all 4 personas

6. Limitations

* Salary data: Kaggle dataset has no structured salary fields. Salary matching relies on description text parsing (~60% coverage). Adzuna API provides structured salary.

* Dataset
Tech-filtered

7. AI Tools

Developed using Claude Code (claude-sonnet-4-6) as the primary development assistant. Full prompt log in prompts.md. AI-generated code reviewed, tested, and modified throughout. Key prompts: system architecture design, multi-stage ranking pipeline, Rocchio adaptive learning, Techmap schema parser, Flask session architecture, and the explainability panel. All benchmark results, persona validation, and architectural decisions reflect original engineering judgment.